Who Gets a Place When Shortlists Get Better?

Evidence from New York rental simulations and Chilean school admissions

ISHAN KHILLAN, Harvard University, USA
ELAINE ZHANG, Harvard University, USA
TODD ZHOU, Harvard University, USA

A recommendation can put the right school on a family's list while making a seat harder to get. We study that distinction by holding preferences and priorities fixed and clearing whole cohorts under alternative shortlists. In 240 ordinary New York rental simulations, a language model showed renters their full-list deferred-acceptance partner 5.62 percentage points more often than uniform menus, while placement fell 2.71 points. In two later Chilean school cohorts, supplying reported rank to an offer predictor covered 2,493 additional historical offers and yielded 985 fewer placements in replay. Both directions had been specified in advance for each school cohort. Of the 3,750 school applicants who newly saw their full-list partner, 3,728 received that exact course. Of the 1,455 who lost a place, 1,004 received exactly the same list under both rules. We give a displacement certificate for people who see their full-list partner yet remain unmatched, and test a coordinated referral to fill vacancies without worsening incumbents' assignments. In the school replays, 224 additional applicants could be placed this way, and only one if referrals had to be among their five highest reported options. Historical school offers are observed. Policy changes in both settings are counterfactual.

1 Introduction

Suppose a school admissions platform can keep five choices on a family's application. With a better predictor, it can more often retain the school that made the family an offer. It is easy to check that improvement one family at a time. Once all the lists compete for seats, a second question arises. Who gets a place?

Consider Chile's 2022 and 2023 regular school application rounds. We applied two predictors trained on 2019 offers to the same submitted preferences. One saw the programs a family listed. The other also saw the order in which the family ranked them. Giving the predictor that order put 2,493 more recorded offers on the lists, including 2,244 more offers that families had explicitly accepted. When we ran applicant-proposing deferred acceptance (DA) on the two sets of lists, 1,455 applicants lost a place and 470 gained one. The historical records establish which offers were covered. The difference of 985 placements comes from rerunning the allocation.

We study the same question from the other direction in New York. There we observe household characteristics and advertised housing, but have to model both sides' rankings. We can hold one inventory fixed, issue five recommendations per renter and observe exactly who competes for which units in the simulation. A second sampling design changes the supply of affordable units and draws lower-income households with an eligible unit in that inventory. In ordinary markets a recommender might improve its average accuracy while doing little for lower-income renters because many have no eligible listing at any price they can pay. The Chilean data let us test whether the prediction–placement tension can arise with submitted preferences and an administrative offer as a target. They do not require the housing preference model to be behaviorally correct.

Our contribution has three parts. First, we compare the same person's change in partner visibility with their change in placement under two market-wide policies. Second, we identify a distinct applicant with an omitted partner for each full-list recipient left unmatched despite seeing their own partner. Third, we compute which unmatched applicants a platform can place by replacing one recommendation with a mutually acceptable vacancy. The result gives both the maximum number of such extensions and a guarantee for each named referral. We report alternative baselines

Authors' Contact Information: Ishan Khillan, Harvard University, Cambridge, Massachusetts, USA; Elaine Zhang, Harvard University, Cambridge, Massachusetts, USA; Todd Zhou, Harvard University, Cambridge, Massachusetts, USA.

throughout. In the Chilean cohorts, both predictors perform better on both outcomes than a uniformly drawn list.

2 What each experiment measures

Let E be the bipartite graph of acceptable applicant–program pairs. A displayed graph $D \subseteq E$ lists the programs an applicant can pursue. Each applicant strictly orders acceptable programs above being unplaced. A program has a fixed capacity and a strict priority ordering over applicants. It holds the highest-priority acceptable applicants who have proposed, up to its capacity. In the rental simulations each listing has one place. Let $M = \mathrm{DA}(E)$ be the full-list allocation and $N_D = \mathrm{DA}(D)$ the policy allocation; an unplaced applicant receives $\perp$. Stability means that no displayed pair would both prefer to match, taking account of vacancies and priorities. DA on a fixed graph is polynomial-time and applicant-optimal among its stable allocations [Gale and Shapley, 1962, Roth, 1986].

For applicant i, define

$$C_i(D) = \mathbf{1}\{M(i) \neq \perp,\ (i, M(i)) \in D\}, \qquad P_i(D) = \mathbf{1}\{N_D(i) \neq \perp\}.$$

Both rates use all applicants. A person without a full-list partner contributes zero to coverage and can still receive a place after the lists change. In Chile we also compute C_i using the historical offer instead of $M(i)$. For policies A and B, the ordered pair $(C_i(B) - C_i(A), P_i(B) - P_i(A))$ belongs to a three-by-three table. The table identifies a person who newly sees their partner yet loses a placement. We use a separate indicator for exact equality of the two personal menus to identify people whose options stayed fixed while competitors' lists changed. Unchanged coverage by itself does not imply that a menu stayed fixed.

Table 1. Two policy comparisons with a common allocation outcome. Each school application cycle is a separate national cohort.

	New York rentals	Chilean school admissions
Choices and priorities	Affordability and space rules, modeled tastes and screening	Submitted course rankings, published priorities and course lotteries
Shortlist change	Uniform five to Haiku five	Rank-blind predictor to rank-informed predictor, five discretionary choices plus entitlement
Reference option	Simulated full-list DA unit	Full-list replay program and recorded initial offer
Evidence	Development simulation, 20 seed blocks per supply design	Frozen 2019 models, two later cohorts for the specified signs

Relation to prior work. Competition between recommended options has a long history. Su et al. [2022] optimize rankings jointly when job seekers apply and employers decide whom to interview. Ashlagi et al. [2020] study recommendations that reduce communication in matching markets. McCauley et al. [2026] link unmatched hospitals under prefix truncation to missed stable partners. Our comparisons permit deletions anywhere on a submitted list and trace who is displaced. We use the classical invariance of matched participants on a fixed graph and results on entry and consistency [Crawford, 1991, Gale and Sotomayor, 1985, Roth, 1986, Sasaki and Toda, 1992].

Aguiar and Hong [2026] compare stable and maximum assignments on restricted graphs and use Chilean school data to study retained rankings. We share their distinction between feasible and stable placements. We compare alternative fitted rules at a fixed quota and identify whose

placement changes without a change in their own menu. Arteaga et al. [2022] show in a field intervention that advice about admission risk can induce families to add options and improve actual placements. Here all comparisons hold the submitted reports fixed. Liu et al. [2024] audit the housing options language models recommend. Our housing exercise follows each recommendation through competition for a particular unit.

3 What does showing a partner guarantee?

The full-list partner in this section is an assignment in the same model as the restricted replay. The Chilean historical offer provides a separate empirical target.

The best prediction can send two people to one seat. Three applicants share three one-seat programs x, y, z. Their first choices, in applicant order, are (x, y, z), (x, z, y), (y, x, z) and (y, z, x) with probabilities $2/6, 2/6, 1/6, 1/6$. Each state has distinct first choices, so all three receive those choices under full-list DA whatever the program priorities. The platform knows the probabilities but cannot see the state. With one slot per applicant, the unique choices maximizing expected partner coverage are (x, z, z). They cover $5/3$ partners in expectation and place two people. The lists (x, y, z) cover $3/2$ partners in expectation and place all three. Thus filling the third seat can cost as little as $1/6$ of an expected correct recommendation in this example.

What an omitted partner can explain. Let H be the applicants whose full-list assignment is a real program missing from D. Let K_j and T_j count members of H assigned to j under M and N_D, respectively.

Theorem 3.1. *Relative to the full market, gross losses and gross gains in placement on D are each at most $|H| - \sum_j \min(K_j, T_j)$. Furthermore, every unmatched applicant whose full-list partner is displayed can be paired with a distinct member of H who is placed on D and strictly prefers their full-list partner. Both conclusions hold for arbitrary stable selections on the two graphs.*

Proof. Delete H together with the seats its members occupy in each assignment. Any remaining blocking pair would have blocked the original assignment. The remaining applicants thus have stable outcomes on the same displayed pairs, using capacities $c - K$ and $c - T$, where c is the original capacity vector. In the first market their full-list partners all remain visible. For fixed pairs, adding one seat retains every matched applicant and adds at most one: regard seats as unit copies with common order within each program, run seat-proposing DA, and admit the new proposer last. An applicant who holds an offer can only trade up. Matched-set invariance transfers this comparison to any stable selection [Crawford, 1991, Gale and Sotomayor, 1985, Roth, 1986]. Comparing the two capacity vectors through their coordinatewise minimum bounds losses among the remaining applicants by $\sum_j (T_j - K_j)_+$ and gains by $\sum_j (K_j - T_j)_+$. Exactly $|H| - \sum_j T_j$ of H lose a place, and none gain. Adding gives the bounds.

For the individual claim, at each program cancel the holders common to both assignments. Pair departing holders with incoming holders one-to-one until one side is exhausted. Start with an applicant who is unmatched on D despite seeing their full-list partner j. Program j must be full in N_D, and every new holder there outranks this applicant. Follow the pairing at j to one of its new holders h. Full-list stability implies that h had an assignment strictly preferred to j, since j previously held a lower-priority applicant. If h still sees that former assignment, repeat. Each step starts with someone who strictly prefers their former assignment and each pairing is injective. On a continued step the program is full after restriction. Its intake of new holders is therefore at least its number of departing holders, so the departing applicant always has a successor. The starting applicant has no incoming link, so the paths cannot cycle or meet. Each path ends at a placed applicant whose former assignment is hidden. □

For two policies A, B, one entry in the empirical comparison counts applicants with $C_i(B) - C_i(A) = 1$ and $P_i(B) - P_i(A) = -1$. The theorem gives each such applicant a distinct other person on the final display who is assigned a worse program than their full-list partner, which is hidden under B. That other applicant can have the same list under A and B. It can happen with only two applicants and one slot each. Let i prefer x to y, let h prefer y to x, and let both programs give h higher priority. Full-list partners are x for i and y for h. Display A offers i only y and h only x. Both place. Change only i's list to x. Now i sees their full-list partner and loses the competition for it. Under one-slot lists every witness is this direct: a new holder at the focal program could not also see their own strictly preferred full-list partner. The traced links in a larger market describe the two assignments and need not follow the chronology of DA proposals.

Let $p_{ij}(X)$ be the probability that j is the full-list partner of i, given the information X used to choose the lists. If a policy also uses randomness independent of the full allocation conditional on X, condition on its seed as well. By linearity, the conditional expected number covered is $\sum_i \sum_{j \in D(i)} p_{ij}(X)$. Selecting the most probable partners separately for each person therefore minimizes expected omissions, the coarse upper bound $|H|$ above. A smaller upper bound can coexist with fewer actual placements. For comparison, McCauley et al. [2026, Theorem 4.7] identify which hospitals can become unmatched under one-sided prefix pruning. Here omissions can occur anywhere in an incomplete list, and a displaced person can still see their own full-list partner.

4 New York City rentals

We paired archived model recommendations with a simulation of who could rent each apartment. The inputs are recent-mover household records from the 2020–24 American Community Survey and rental advertisements first listed in June–August 2026 [FirstMover, 2026, U.S. Census Bureau, 2026]. Each market contains 64 renter instances and 64 available apartments from one borough and month. An apartment must meet bedroom needs and cost at most 40% of monthly income. Lower-income households earn at most the survey-weighted median for NYC cash renters. Renter preferences combine rent, size, physical features, distance and taste shocks. Listing priorities depend on income relative to rent and a random shock specific to the renter–listing pair. We rerun applicant-proposing DA with exactly the same households, apartments and preferences under each display. The resulting placements are simulated. The ads provide asking rents, and the survey describes renter households. Neither source observes offers or leases between these parties.

In the ordinary design we sample households using survey weights and draw available apartments uniformly. In the enriched design half the inventory comes from a borough-specific affordability band. We then sample 32 lower-income households with a feasible option in that inventory and 32 higher-income households. This deliberate change lets us examine competition where lower-income households can participate. Each design has 20 seed blocks across three months and four supported boroughs, giving 240 paired markets and 15,360 renter instances. Staten Island had too few listings for draws of 64. The supported boroughs represent 97.95% of the intended recent-mover geographic weight.

The income cutoff is $64,435 in 2024 dollars. In asking-rent dollars at the study's July 2026 price base, the highest monthly rent allowed at the cutoff is $2,286. We use the calibrated preference specification from the rental pilot. It selects a rent coefficient of 4 and an idiosyncratic shock scale of 0.5 from a nine-point grid. Its burden calibration has an observable limitation. At 217 lower-income placements across the 240 ordinary calibration markets, even a feasible assignment chosen to minimize their first-quartile rent burden cannot push that quartile below 27.78%. The target quartile from recent movers' contract rents was 22.00%. These finite-market bounds concern the calibration exercise. The household survey does not validate the assumed listing priorities or individual rankings.

Table 2. NYC. Change under Haiku relative to uniform five-listing menus, in percentage points, with paired 95% simulation intervals. Both columns use every renter in the row. A renter without a full-list DA partner contributes zero to partner coverage in both policies.

Design	Renters	Full-list partner shown	Placed by DA
Ordinary	All	+5.62 [4.96, 6.27]	−2.71 [−3.19, −2.23]
	Lower income	+0.11 [0.00, 0.24]	+0.00 [−0.19, 0.21]
	Higher income	+9.17 [7.99, 10.34]	−4.58 [−5.27, −3.86]
Enriched	All	+5.85 [5.04, 6.60]	−5.40 [−6.05, −4.73]
	Lower income	+0.78 [0.19, 1.43]	−0.92 [−2.23, 0.37]
	Higher income	+10.92 [9.47, 12.29]	−9.88 [−10.43, −9.36]

What the recommender sees. For each renter, the five-slot uniform rule draws from the feasible listings without replacement. The archived Claude Haiku 5.5 recommender receives the same candidate set, a budget, bedroom needs, target area, prices, physical features and distances. The prompt asks it to choose five listings. It has no observations of other renters' menus or the realized preferences. Both policies show all feasible options to anyone with five or fewer. We used the 20,740 saved valid responses from the development pilot. Households and apartments can recur across draws. Rates below pool individual indicators within each borough-month and then average cells with equal month weights and Census recent-mover borough weights. The pointwise intervals resample entire paired seed blocks 10,000 times. They condition on the catalogues, behavioral model and stored replies.

What changed when everyone used the model? Under the original pilot design the primary endpoint was the lower-minus-higher difference in hidden blocking pairs. The comparison here follows the same people and uses the saved menus. Ordinary markets placed 53.89% of renters under uniform lists and 51.18% under Haiku, even as full-list partner coverage rose from 9.42% to 15.04% (Table 2). The direction depends on the comparator. Against a fixed score that ranks eligible apartments using their observable features, Haiku placed 20.26 points more renters in ordinary markets and 24.47 points more in enriched markets. Its coverage differences were −0.44 and +0.66 points, respectively.

What the group averages conceal. In the ordinary design, 5,425 of 6,617 lower-income renter instances had no feasible listing. The standardized change in their placement rate was smaller than one hundredth of a point. In enriched markets the change was −0.92 points with an interval spanning zero, yet 1,203 lower-income instances lost a placement and 1,044 gained one. The enriched lower-income share with a hidden blocking pair (a renter and an unshown apartment that each prefer the other to their assignment) rose from 37.03% to 41.31%. The higher-income share fell from 85.19% to 78.01%. The absolute gap narrowed even as hidden blocking pairs became more common for the lower-income group. Partner coverage and placement were secondary outcomes in this development pilot. We treat these results as evidence of a mechanism under specified assumptions.

5 Chilean school admissions

The Chilean school choice system records the courses a family ranks, the school's priority and lottery information, and the initial offer the family actually receives [Correa et al., 2022, Ministerio de Educación de Chile, 2024]. We can ask whether a hypothetical shortlist would have included that offer. To learn who would receive a place on the new lists, we must replay the allocation.

We fitted two ridge logistic offer predictors to the same 483,070 applicants in 2019, using each eligible applicant–course edge as an observation. Both use program demand, capacity, congestion,

list length, grade, priority indicators and a smoothed program offer rate estimated from 2019 (leave-one-out for training). The second adds log rank, a first-choice indicator and a rank–congestion interaction. Neither policy sees the realized offer. The lottery is used for allocation only. Each predictor retains up to five discretionary choices. A pupil's guaranteed current-school option remains available outside that quota. The selected courses keep their submitted order when applicants propose. We use the same reported priorities, realized course lotteries and pooled capacities for both replays. Full-list DA reproduces the recorded program or outside option for 95.39% of applicants in 2022 and 95.79% in 2023.

A test in two later cohorts. After examining the 2020 and 2021 application years, we recorded a joint prediction before obtaining the 2022 and 2023 releases. These application years correspond to school entry in 2023 and 2024. The cohorts include 520,420 and 492,888 initial offer recipients, respectively. All applicants enter the placement and joint-transition counts. Giving the 2019 predictor access to rank would cover more historical initial offers and reduce replayed placements in each subsequent cohort. Table 3 reports both outcomes, with full-list DA partners as a second, directly comparable coverage target. Among the 2,493 net additional historical offers covered in the two years, 2,244 were offers that families explicitly accepted. The historical record does not tell us whether they would have accepted under either new policy.

Table 3. Rank-informed minus rank-blind school menus. Placement and full-list DA coverage come from the replay. Historical offer coverage uses the recorded initial course. Each entry is a net count in the same application-year cohort.

Year	Applicants	Historical offers shown	Full-list partners shown	Placements
2022	570,891	+1, 143	+1, 261	−468
2023	536,353	+1, 350	+1, 473	−517

The choice of comparison matters. In a follow-up we reran DA on 48 uniform five-choice displays in each cohort and on the family's first five choices. Relative to the uniform-draw mean, the rank-informed policy covered about 9,235 and 9,740 more full-list partners and placed about 789 and 759 more pupils in 2022 and 2023. Relative to first-five, it placed 2,123 and 2,013 more. The rank-blind predictor placed the most pupils of these four policies in both years. Prioritizing predicted offers improved on simple lists, and adding rank information then moved the two outcomes in opposite directions.

Pooling all seats simplifies the ministry's allocation. A sensitivity that first honors continuity and gives designated pupils priority for earmarked vacancies changes the year-specific net placement losses to 469 and 516. Neither version models all reserves or dynamic family priorities. These are whole-cohort comparisons under fixed reports and a fixed lottery. The new transition counts and comparisons with uniform are exploratory. The joint directional test involved only the recorded-offer and placement outcomes under the two frozen fitted policies.

6 Who loses a placement?

At the level of a whole market, both studies give instances of higher coverage and fewer placements. The losses reach different people. Figure 1 partitions everyone who was placed under A and unplaced under B. We first separate people whose menu was identical. Among the rest, we ask whether their full-list partner became visible, became hidden, or had the same visibility as before.

A change that reaches someone else's list. In Chile, only 32,914 of 1,107,244 application records had different displayed courses under the two predictors. Of 1,455 applicants who lost a placement,

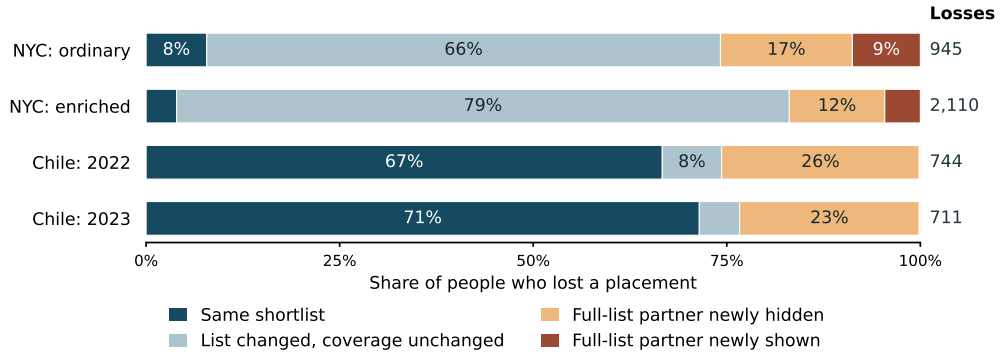


Fig. 1. What changed for people who lost a placement? Bars within each row sum to 100%. Numbers at right are raw gross losses. A changed list can retain the same partner coverage, including when the applicant had no full-list partner. The latter applicants still count in the analysis.

1,004 had exactly the same personal menu. The reverse happened to 406 of the 470 who gained one. Holding preferences, priorities, capacities and an applicant's own list fixed, a change in that applicant's placement can arise only through competitors' menus. These spillovers account for a net loss of 598 seats among applicants whose lists stayed fixed. Among the 32,914 applicants with edited lists, 451 lost a place (1.37%). Among the 1,074,330 with identical lists, the 1,004 losses represent 0.093%.

In the ordinary NYC markets, 945 renter instances lost a placement and 475 gained one. Of the 945 who lost, 74 saw identical menus and 83 newly saw their full-list partner. In enriched markets, 2,110 lost a placement and 1,212 gained one. Here 83 of those who lost saw identical menus and 97 newly saw their partner. Over both designs, newly seeing one's full-list partner and losing a place happened to 180 renter instances. For all 180 we found the distinct displaced-partner witnesses in Theorem 3.1 within their respective market. Of these paths 169 reached a witness after one program. Longer paths involved up to three programs.

The newly shown school offer often converted. In Chile, 3,750 applicants newly saw their full-list DA partner under the rank-informed predictor. Of these, 3,728 obtained that exact course in the replay and 20 obtained a different course. The remaining two went unplaced. In ordinary NYC markets 1,056 of the 1,720 newly shown renters obtained the named apartment (61.4%); in enriched markets it was 1,277 of 2,012 (63.5%). Across all applicants in each NYC design the number receiving their exact full-list partner increased by 523 and 553, respectively. Across the two school cohorts it increased by 5,049. More people therefore received their exact full-list partner even while fewer received any placement. We verified both displacement paths. Analyzing the recorded historical offer as a separate target, 3,366 of the 3,622 applicants who newly saw that program received it in the replay. Thus a platform could document a high rate of placement at the newly shown program and still miss a loss elsewhere in the market. The witness theorem applies to the modeled full-list partner. It uses stability and cannot be applied to an arbitrary historical offer.

Feasibility and priorities. Let $v(D)$ be the largest number who can simultaneously take displayed options within capacity, and $g(D) = v(D) - |N_D|$ the feasible places unrealized by DA. The placement change is $\Delta v - \Delta g$ [Aguiar and Hong, 2026]. In ordinary NYC markets, displayed capacity fell by 3.84 percentage points and the shortfall shrank by 1.13, giving the observed 2.71-point decline. In

enriched markets those changes were −12.29 and −6.89 points. The school results differ. In 2022 the display could accommodate 244 fewer applicants and the shortfall grew by 224, for a loss of 468. In 2023 those figures were 153 and 364, for a loss of 517. Each school maximum is certified by a feasible flow and a cut of equal value. This benchmark uses only displayed edges and seat counts. It relaxes priorities and the obligation to place a particular returning pupil. These identities describe two margins on the same graph. They do not isolate the causal effect of changing priorities.

7 How many vacancies could one referral fill?

Once the platform has tentative offers, it knows which people and seats are still available. Suppose it can contact an unmatched applicant with one currently vacant option. A vacancy alone says little about who can use it. An available apartment may be outside the renter's budget, and an available school may be one the family never listed. In this section each referral must use a pair already present in the feasible graph. The calculation is an exploratory follow-up to both experiments.

7.1 One coordinated referral after offers

Suppose the platform now observes an assignment M_0 stable on the initial lists. Join each unmatched applicant to each mutually acceptable program with a current vacancy. These edges are hidden: if one were displayed it would block M_0. Give each program its remaining capacity and let r be the size of a maximum matching in this graph. With lists of $\min(k, d_i)$ programs, where $k \geq 1$ and d_i is feasible degree, a hidden option can always replace a displayed one. If an entitlement is protected outside the quota, the substitution is made in a discretionary slot. In both replays, every selected applicant had a full five-slot discretionary list.

Theorem 7.1. *The largest number of additions compatible with retaining every initial contract is r. One shortlist substitution per selected applicant attains it. Every stable outcome on the revised display gives each consenting applicant their named referral, even if others decline. Under applicant-proposing DA, every original recipient also receives a weakly preferred assignment.*

Proof. The new contracts in any extension must form a residual matching. Conversely, choose any such matching A. For each consenting participant substitute their assigned program for their last discretionary displayed option. Write W for the original assignment together with the accepted referrals. All new pairs in the display belong to W. On old edges the applicant weakly improved. An originally full program has the same holders. At an originally slack one, every applicant with an old edge weakly preferred their original assignment, by stability. Hence W is stable on this display. DA weakly dominates W for applicants. All stable selections have the same matched people [Roth, 1986]. If some stable selection omitted an accepted referral e, it would keep its recipient matched even after deleting that unused edge. On that same graph $W \setminus \{e\}$ is stable by the preceding argument, since this applicant now has a subset of their original list. That would leave the person unmatched in one stable outcome only, a contradiction. □

The maximum fixes the original contracts in a stable witness. Opening new options to incumbents would be a different intervention. The empirical calculation also requires deciding which hidden options a person would actually accept. We report both a top-five rank limit and all mutually acceptable vacancies. Work on facilitating matches, adding school seats and guaranteed school referrals provides related interventions [Gajulapalli et al., 2020, Lepe, 2025, Trabelsi et al., 2025].

The NYC totals mask a distributional constraint. In 240 ordinary markets only 37 initially unplaced lower-income renter instances could use any vacancy. Among all ways of achieving the maximum of 683 additional placements, between 22 and 25 could go to them. In enriched markets 1,601 initially unplaced lower-income renters could use a vacancy. At a total of 1,926 additions, between

Table 4. Largest number of named referrals compatible with retaining all existing contracts in one stable allocation. NYC uses simulated Haiku offers. Chile uses rank-informed pooled DA offers. All counts use instances in their own row. A vacancy must be at an affordable, feasible listing or an eligible course already on the pupil's reported list.

	Vacant seats	Unplaced with an option	Any rank	Top five
NYC ordinary	7,972	757	683	204
NYC enriched	5,990	2,527	1,926	370
Chile 2022	601,565	128	127	0
Chile 2023	640,751	97	97	1

1,002 and 1,085 could go to the lower-income group. We obtained the endpoints by maximizing cardinality and then minimizing or maximizing that group's count. The enriched design changes both supply and who is drawn. The difference between the rows is a scenario comparison.

In Chile the seat totals sum all residual capacities in the course-level replay, including courses that no unmatched applicant had listed. At most 224 placements can be added over the two application years by this contract-preserving intervention, assuming full take-up of an eligible referral. One applicant could be helped within the family's top five. The other available referrals require going further down the original list. In the rank-blind comparison there were zero eligible referrals in both years. After giving each policy its own largest extension, rank-informed still placed 761 fewer applicants in total. In New York, starting from uniform lists instead gave maxima of 215 ordinary and 518 enriched placements. The number of available second-round referrals depends in part on what the first round already achieved.

A second implementation solved the school flow problems on each entire national graph and independently replayed its assignments. For every selected applicant we replaced the last discretionary option with the named listing or course and reran DA. We checked that every referral was honored, that all earlier recipients kept a weakly preferred assignment and that each list kept its original size. We preserved school continuity rights throughout. An operational service would have to elicit acceptable alternatives, observe current offers, and offer a real opportunity to decline. The counts in Table 4 measure opportunities under those assumptions. They do not measure actual take-up.

8 Implications and limits

Consider an evaluator who checks only whether the eventual offer appeared among the recommendations. In the school replays that evaluator would endorse adding rank information. Checking only applicants whose lists changed would still miss roughly seven in ten gross placement losses. A different evaluator who compared the same rank-informed model with uniform lists would see improvements on both measures. All three observations can hold for the same students and capacities. For a decision about a shortlist change, the comparator is the policy the platform would actually replace.

The rental experiment addresses a different social concern. A renter who cannot afford a single listing has zero hidden blocking pairs and zero chance of placement under either policy. A group can consequently score well on a hidden-options measure while having almost no access to housing. We keep these households in the denominator, report their feasible sets, and examine how much discretion the platform has once tentative offers exist. Giving lower-income renters priority among equally large referral assignments had room to move only three additional placements toward them

in the ordinary draws. A service that also changes affordability or expands search would face a different feasible graph.

Limits of the evidence. The housing data identify asking rents and surveyed household characteristics. Preferences, housing-provider priorities, budgets and assignments are modeled. The budget in the prompt is correlated with income, even though group membership is absent from the prompt. Results condition on one model version, one stored response per queried renter and a small number of independent simulation blocks. In Chile the rank-informed model already has access to families' ordered applications. Its evidence concerns curation of known choices. The actual admissions system uses reserves and family contingencies beyond the pooled-priority replay. Changing what a platform displays could change what people search for, how they report, who applies, or whether they accept. Neither study estimates those responses or a causal effect of deployment. The two Chilean application years may include some of the same children. We treat them as two finite cohorts and report exact counts. The two settings also have different treatments, so their effects have no common causal scale.

What a platform can audit. Record the available options and capacities, the two complete sets of proposed lists, the reference offer where one exists, and the assignment induced by each set of lists. Those records distinguish a successful recommendation to its intended recipient, a lost place on a list that never changed, and a person without any feasible alternative. The last case points to a constraint on supply. The first two call for evaluating recommendation and allocation together.

Reproducibility. The accompanying code regenerates the two studies' aggregate tables and figure. In NYC we checked 2,880 published market-by-group endpoint rows against the regenerated menus. The legacy reply cache stores listing indices without a digest of the original prompt payload. Its consistency with this reconstruction can be checked, while the exact original payload cannot be authenticated independently. In Chile we recomputed the blind, rank-informed and full-list assignments in both cohorts and checked all 3,321,732 applicant–policy outcomes against the archived serial solver. The two-year sign criterion is documented in the team's local study record before data acquisition. The joint transitions, alternative baselines in the later years, and referral analyses are follow-ups. All shared school outputs are aggregates.

References

Victor H. Aguiar and Dian Hong. 2026. The Attention Cost of Stable Matching. arXiv:2609.09693. https://doi.org/10.48550/arXiv.2609.09693

Felipe Arteaga, Adam J. Kapor, Christopher A. Neilson, and Seth D. Zimmerman. 2022. Smart Matching Platforms and Heterogeneous Beliefs in Centralized School Choice. *Quarterly Journal of Economics* 137, 3 (2022), 1791–1848. https://doi.org/10.1093/qje/qjac013

Itai Ashlagi, Mark Braverman, Yash Kanoria, and Peng Shi. 2020. Clearing Matching Markets Efficiently: Informative Signals and Match Recommendations. *Management Science* (2020). https://doi.org/10.1287/mnsc.2018.3265

José Correa, Natalie Epstein, Rafael Epstein, Juan Escobar, Ignacio Rios, Nicolás Aramayo, Bastián Bahamondes, Carlos Bonet, Martin Castillo, Andrés Cristi, Boris Epstein, and Felipe Subiabre. 2022. School Choice in Chile. *Operations Research* 70, 2 (2022), 1066–1087. https://doi.org/10.1287/opre.2021.2184

Vincent P. Crawford. 1991. Comparative Statics in Matching Markets. *Journal of Economic Theory* 54, 2 (1991), 389–400. https://doi.org/10.1016/0022-0531(91)90129-R

FirstMover. 2026. NYC Rental Listings at First Appearance. FirstMover Open Data Project. Monthly public releases.

Karthik Gajulapalli, James A. Liu, Tung Mai, and Vijay V. Vazirani. 2020. Stability-Preserving, Time-Efficient Mechanisms for School Choice in Two Rounds. In *40th IARCS Annual Conference on Foundations of Software Technology and Theoretical Computer Science*. https://doi.org/10.4230/LIPIcs.FSTTCS.2020.21

David Gale and Lloyd S. Shapley. 1962. College Admissions and the Stability of Marriage. *The American Mathematical Monthly* 69, 1 (1962), 9–15. https://doi.org/10.1080/00029890.1962.11989827

David Gale and Marilda Sotomayor. 1985. Some Remarks on the Stable Matching Problem. *Discrete Applied Mathematics* 11, 3 (1985), 223–232. https://doi.org/10.1016/0166-218X(85)90074-5

Ignacio Lepe. 2025. *Designing Smart School Choice Recommendations: Heuristics for Sure Alternatives.* Working paper 2. ConsiliumBots. https://consiliumbots.github.io/working-papers-cb/L_2025/working_paper/L_2025.pdf

Eric Justin Liu, Wonyoung So, Peko Hosoi, and Catherine D'Ignazio. 2024. Racial Steering by Large Language Models: A Prospective Audit of GPT-4 on Housing Recommendations. In *Proceedings of the 4th ACM Conference on Equity and Access in Algorithms, Mechanisms, and Optimization.* Association for Computing Machinery. https://doi.org/10.1145/3689904.3694709

Samuel McCauley, Benjamin Moseley, Helia Niaparast, and Shikha Singh. 2026. Stable Matching with Predictions: Robustness and Efficiency under Pruned Preferences. arXiv:2602.02254. https://doi.org/10.48550/arXiv.2602.02254

Ministerio de Educación de Chile. 2024. Sistema de Admisión Escolar (SAE): Postulaciones, oferta y resultados. https://datosabiertos.mineduc.cl/sistema-de-admision-escolar-sae/ Public regular-round releases for application years 2019–2023.

Alvin E. Roth. 1986. On the Allocation of Residents to Rural Hospitals: A General Property of Two-Sided Matching Markets. *Econometrica* 54, 2 (1986), 425–427.

Hiroo Sasaki and Manabu Toda. 1992. Consistency and Characterization of the Core of Two-Sided Matching Problems. *Journal of Economic Theory* 56, 1 (1992), 218–227. https://doi.org/10.1016/0022-0531(92)90078-V

Yi Su, Magd Bayoumi, and Thorsten Joachims. 2022. Optimizing Rankings for Recommendation in Matching Markets. In *Proceedings of the ACM Web Conference.* https://doi.org/10.1145/3485447.3511961

Yohai Trabelsi, Abhijin Adiga, Yonatan Aumann, Sarit Kraus, and S. S. Ravi. 2025. Facilitating Matches on Allocation Platforms. In *Proceedings of the 28th European Conference on Artificial Intelligence.* https://doi.org/10.3233/FAIA251269

U.S. Census Bureau. 2026. American Community Survey 2020–2024 5-Year PUMS User Guide and Overview. American Community Survey Office.